```
!pip install rdkit-pypi
!pip install pandas numpy tensorflow
```

Show hidden output

```
from google.colab import files
uploaded = files.upload()
```

Choose Files 250k_rndm...clean_3.csv

250k_rndm_zinc_drugs_clean_3.csv(text/csv) - 22606589 bytes, last modified: 4/16/2026 - 100% done

```
Saving 250k_rndm_zinc_drugs_clean_3.csv to 250k_rndm_zinc_drugs_clean_3.csv
```

```
import pandas as pd

df = pd.read_csv('250k_rndm_zinc_drugs_clean_3.csv')
smiles = df['smiles'].dropna().tolist()

print(len(smiles))
```

```
249455
```

```
!pip install rdkit
from rdkit import Chem

def is_valid(s):
    return Chem.MolFromSmiles(s) is not None

smiles = [s for s in smiles if is_valid(s)]
```

```
Collecting rdkit
  Downloading rdkit-2026.3.1-cp312-cp312-manylinux_2_28_x86_64.whl.metadata (3.8 kB)
Requirement already satisfied: numpy in /usr/local/lib/python3.12/dist-packages (from rdkit) (2.0.2)
Requirement already satisfied: Pillow in /usr/local/lib/python3.12/dist-packages (from rdkit) (11.3.0)
Downloading rdkit-2026.3.1-cp312-cp312-manylinux_2_28_x86_64.whl (37.0 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 37.0/37.0 MB 63.0 MB/s eta 0:00:00
Installing collected packages: rdkit
Successfully installed rdkit-2026.3.1
```

```
PAD = '_'

chars = sorted(list(set(''.join(smiles)) | {PAD}))
char_to_idx = {c:i for i,c in enumerate(chars)}
idx_to_char = {i:c for c,i in char_to_idx.items()}
```

```
max_len = 30
batch_size = 256
```

```
import numpy as np

def data_generator(smiles, batch_size):
    while True:
        X_batch = []
        y_batch = []

        for s in smiles:
            for i in range(min(len(s)-1, 10)):  # limit expansion
                seq = s[:i+1].ljust(max_len, PAD)
                X_batch.append([char_to_idx[c] for c in seq])
                y_batch.append(char_to_idx[s[i+1]])

                if len(X_batch) == batch_size:
                    yield np.array(X_batch), np.array(y_batch)
                    X_batch, y_batch = [], []
```

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense, Embedding

model = Sequential([
    Embedding(len(chars), 64, input_length=max_len),
    LSTM(128),
    Dense(len(chars), activation='softmax')
])

model.compile(loss='sparse_categorical_crossentropy', optimizer='adam')
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/embedding.py:100: UserWarning: Argument `input_length` is depr
  warnings.warn(
```

```
steps_per_epoch = len(smiles) // batch_size

model.fit(
    data_generator(smiles, batch_size),
    steps_per_epoch=steps_per_epoch,
    epochs=50
)
```

```
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7883
Epoch 23/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7900
Epoch 24/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7904
Epoch 25/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7899
Epoch 26/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7861
Epoch 27/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7868
Epoch 28/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7826
Epoch 29/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7837
Epoch 30/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 0.7829
Epoch 31/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7816
Epoch 32/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7796
Epoch 33/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7799
Epoch 34/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7823
Epoch 35/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 0.7791
Epoch 36/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7783
Epoch 37/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7794
Epoch 38/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7750
Epoch 39/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7764
Epoch 40/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7756
Epoch 41/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7757
Epoch 42/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7734
Epoch 43/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 6ms/step - loss: 0.7734
Epoch 44/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7763
Epoch 45/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 0.7738
Epoch 46/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7738
Epoch 47/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7741
Epoch 48/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7701
Epoch 49/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 7s 7ms/step - loss: 0.7714
Epoch 50/50
974/974 ━━━━━━━━━━━━━━━━━━━━ 6s 7ms/step - loss: 0.7718
<keras.src.callbacks.history.History at 0x7e162695b6b0>
```

```
model.save("molecule_model.h5")
```

```
WARNING:absl:You are saving your model as an HDF5 file via `model.save()` or `keras.saving.save_model(model)`. This file for
```

```
allowed_chars = set(char_to_idx.keys())

def clean_smiles(sm):
    return ''.join([c for c in sm if c in allowed_chars])
```

```
import numpy as np


# Sampling function (stable)
def sample(preds, temperature=0.3):
    preds = np.log(preds + 1e-8) / temperature
    preds = np.exp(preds)
```

```
    preds = preds / np.sum(preds)
    return np.random.choice(len(preds), p=preds)




# Generate SMILES (with strict control)
def generate_smiles(seed, length=30):
    s = seed


    for _ in range(length):
        seq = s[:max_len].ljust(max_len, PAD)
        x = np.array([[char_to_idx.get(c, 0) for c in seq]])


        preds = model.predict(x, verbose=0)[0]


        next_idx = sample(preds, temperature=0.3)
        next_char = idx_to_char[next_idx]


        # STOP CONDITIONS (VERY IMPORTANT)
        if next_char == PAD:
            break


        if next_char not in char_to_idx:
            break


        if next_char in ['[', ']', '@', '/', '\\']:
            break


        s += next_char


        if len(s) > max_len:
            break


    return s




# CLEAN FUNCTION
allowed_chars = set(char_to_idx.keys())


def clean_smiles(sm):
    return ''.join([c for c in sm if c in allowed_chars])




# SAFE SEEDS
seeds = ["CC", "CCC", "COC", "CCO"]


# GENERATE MOLECULES
generated = [clean_smiles(generate_smiles(seed=np.random.choice(seeds))) for _ in range(500)]


# REMOVE SHORT + DUPLICATES
generated = [g for g in generated if len(g) > 5]
generated = list(set(generated))


# PRINT OUTPUT
print("Generated Molecules:")
for g in generated:
    print(g)
```

```
CC(C)(C)OC(=O)c1ccc(C(=O)Nc1n))
CCCNC(=O)c1ccc(Cl)cc(C(=O)N3C
CCC(=O)N1CCN(C(=O)
CCCCN1C(=O)
CCCNC(=O)c1cccc(C(=O)Nc2c1c
CCOc1ccc(C(=O)Nc2ccc(NC(=O)(
CC(C)(C)OC(=O)C1=C(C)(C)
COC(=O)c1ccc(C(=O)Nc2cc(C(=H
COC(=O)c1cc(C(=O)Nc2cnc1N1=3C34
COC(=O)C1=C(C(=O)N2c(C)c1nn33PP
CCCCCCNC(=O)N1CCC
CC(=O)N1CC
CC(C)CCNC(=O)NC(=O)c1nnc1C1N
COCCN1C(=O)C(=O)N(c2cc3c
COC(=O)C1=C(C)=C(C)
COC(=O)c1ccc(C(=O)Nc2cc(C(=HH
CCCCNC(=O)CN1C(=O)
CCOc1ccc(NC(=O)c2ccc(Cl)n1CCCH4
CCOC(=O)C1=C(C)=C(N)
COC(=O)C1=C(C(=O)N2c(N+
CCOc1ccc(C(=O)NC(=O)c2cnn1CC134
CCCCN(C)C(=O)NC(=O)c1nn1CC1C344
CCOc1ccc(NC(=O)c2ccc(
CCOc1ccc(C(=O)N
CCOC(=O)c1ccc(Cl)cc(CN2CCN3C3
COCCN1CC
COC(=O)C1=C(C(=O)N2c(C)c1-nns


CC(C)Cn1c(C)c(Cl)ccc(C(=O)N3C
CCC(=O)N1CCC
COC(=O)c1ccc(Cl)cc(C(=O)Nc1n134
CC(C)(C)OC(=O)N1CCC
CCCOc1ccc(C(=O)NNC(=O)N
COC(=O)C1=C
CCOC(=O)C1=C(C)(
CCCCNC(=O)NC(=O)C1CCN(c2c
CCCCNC(=O)c1cccc(NC(=O)(=O)c1nP
CCOc1ccc(C(=O)Nc2cc(C(=O)Nc1n)P
CCCN(CC(=O)N1CCC
CCCC(=O)N1CC
CCOC(=O)c1ccc(Cl)cc(NC(=O)N1C
CC(C)CCNC(=O)NC(=O)N1C=C23
COC(=O)c1cccc(C(=O)Nc2cc(C(
CCCCN(C)C(=O)NC(=O)C1CC1N3C33SI
CCOC(=O)C1=C(C)
CCOc1ccc(CNC(=O)Nc2ccc(C(O
CCOc1ccccc1CN1CN1CCC1CN1C3C33S3
COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP
CC(C)CCOC
CCCCNC(=O)C1CCN(c2ccc3c
CCOc1ccc(
```

```
from rdkit import Chem


valid_raw = []

for sm in generated:
    mol = Chem.MolFromSmiles(sm)
    if mol is not None:
        valid_raw.append(sm)


valid_molecules = list(set(valid_raw))
valid_molecules = [m for m in valid_molecules if len(m) > 5]

print("Valid molecules:", len(valid_molecules))
print(valid_molecules)
```

```
[14:07:18] SMILES Parse Error: Failed parsing SMILES 'COC(=O)c1cccc(C(=O)Nc2cc(C(' for input: 'COC(=O)c1cccc(C(=O)Nc2cc(C(
[14:07:18] SMILES Parse Error: ring closure 3 duplicates bond between atom 14 and atom 15 for input: 'CCCCN(C)C(=O)NC(=O)C
[14:07:18] SMILES Parse Error: unclosed ring for input: 'CCOC(=O)C1=C(C)'
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: CCOc1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] SMILES Parse Error: check for mistakes around position 9:
[14:07:18] CCOc1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] ~~~~~~~~^
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: CCOc1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] SMILES Parse Error: check for mistakes around position 23:
[14:07:18] Oc1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] ~~~~~~~~~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: CCOc1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] SMILES Parse Error: check for mistakes around position 25:
[14:07:18] 1ccc(CNC(=O)Nc2ccc(C(O
[14:07:18] ~~~~~~~~~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: Failed parsing SMILES 'CCOc1ccc(CNC(=O)Nc2ccc(C(O' for input: 'CCOc1ccc(CNC(=O)Nc2ccc(C(O'
[14:07:18] SMILES Parse Error: ring closure 3 duplicates bond between atom 18 and atom 19 for input: 'CCOc1ccccc1CN1CN1CCC
[14:07:18] SMILES Parse Error: syntax error while parsing: COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] SMILES Parse Error: check for mistakes around position 28:
[14:07:18] c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] ~~~~~~~~~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] SMILES Parse Error: check for mistakes around position 13:
[14:07:18] COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] ~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] SMILES Parse Error: check for mistakes around position 24:
[14:07:18] (=O)c1ccc(C(=O)Nc2cc(C(=HHHP
[14:07:18] ~~~~~~~~~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: Failed parsing SMILES 'COC(=O)c1ccc(C(=O)Nc2cc(C(=HHHP' for input: 'COC(=O)c1ccc(C(=O)Nc2cc
[14:07:18] SMILES Parse Error: extra open parentheses while parsing: CCCCNC(=O)C1CCN(c2ccc3c
[14:07:18] SMILES Parse Error: check for mistakes around position 16:
[14:07:18] CCCCNC(=O)C1CCN(c2ccc3c
[14:07:18] ~~~~~~~~~~~~~~~^
[14:07:18] SMILES Parse Error: Failed parsing SMILES 'CCCCNC(=O)C1CCN(c2ccc3c' for input: 'CCCCNC(=O)C1CCN(c2ccc3c'
[14:07:18] SMILES Parse Error: syntax error while parsing: CCOc1ccc(
[14:07:18] SMILES Parse Error: check for mistakes around position 9:
[14:07:18] CCOc1ccc(
[14:07:18] ~~~~~~~~^
[14:07:18] SMILES Parse Error: Failed parsing SMILES 'CCOc1ccc(' for input: 'CCOc1ccc('
```

```
from rdkit import Chem
from rdkit.Chem import Descriptors

def drug_score(sm):
    mol = Chem.MolFromSmiles(sm)

    mw = Descriptors.MolWt(mol)
    logp = Descriptors.MolLogP(mol)
    h_donors = Descriptors.NumHDonors(mol)
    h_acceptors = Descriptors.NumHAcceptors(mol)

    score = 0
    if mw < 500: score += 1
    if logp < 5: score += 1
    if h_donors <= 5: score += 1
    if h_acceptors <= 10: score += 1

    return score / 4
```

```
def toxicity(sm):
    return len(sm) / 50   # simple proxy
```

```
results = []

for sm in valid_molecules:
    d = drug_score(sm)
    t = toxicity(sm)

    score = d - t

    results.append((sm, d, t, score))

# Sort by best score
results = sorted(results, key=lambda x: x[3], reverse=True)

# Print results
print("Final Ranked Molecules:\n")

for sm, d, t, s in results:
    print(f"Molecule: {sm}")
    print(f"Drug Score: {d:.2f}, Toxicity: {t:.2f}, Final Score: {s:.2f}")
    print("-"*40)
```

```
Final Ranked Molecules:

Molecule: CC(C)C
Drug Score: 1.00, Toxicity: 0.12, Final Score: 0.88
----------------------------------------
Molecule: COC(=O)
Drug Score: 1.00, Toxicity: 0.14, Final Score: 0.86
----------------------------------------
Molecule: CC(C)CC
Drug Score: 1.00, Toxicity: 0.14, Final Score: 0.86
----------------------------------------
Molecule: CC(C)(C)
Drug Score: 1.00, Toxicity: 0.16, Final Score: 0.84
----------------------------------------
Molecule: CCOC(=O)
Drug Score: 1.00, Toxicity: 0.16, Final Score: 0.84
----------------------------------------
Molecule: CC(C)CCOC
Drug Score: 1.00, Toxicity: 0.18, Final Score: 0.82
----------------------------------------
Molecule: CCCCNC(=O)
Drug Score: 1.00, Toxicity: 0.20, Final Score: 0.80
----------------------------------------
Molecule: CCCCCCNC(=O)
Drug Score: 1.00, Toxicity: 0.24, Final Score: 0.76
----------------------------------------
Molecule: CCOc1ccccc1C
Drug Score: 1.00, Toxicity: 0.24, Final Score: 0.76
----------------------------------------
Molecule: CC(C)(C)OC(=O)
Drug Score: 1.00, Toxicity: 0.28, Final Score: 0.72
----------------------------------------
Molecule: COC(=O)c1ccccc1C
Drug Score: 1.00, Toxicity: 0.32, Final Score: 0.68
----------------------------------------
Molecule: CCCCNC(=O)NC(=O)
Drug Score: 1.00, Toxicity: 0.32, Final Score: 0.68
----------------------------------------
Molecule: CCOC(=O)c1ccccc1C
Drug Score: 1.00, Toxicity: 0.34, Final Score: 0.66
----------------------------------------
Molecule: CCCCCNC(=O)NC(=O)
Drug Score: 1.00, Toxicity: 0.34, Final Score: 0.66
----------------------------------------
Molecule: CCCCCCNC(=O)NC(=O)
Drug Score: 1.00, Toxicity: 0.36, Final Score: 0.64
----------------------------------------
Molecule: CCCCN(C)C(=O)NC(=O)
Drug Score: 1.00, Toxicity: 0.38, Final Score: 0.62
----------------------------------------
Molecule: COC(=O)c1ccccc1NC(=O)
Drug Score: 1.00, Toxicity: 0.42, Final Score: 0.58
----------------------------------------
Molecule: CCCCCNC(=O)CN1C(=O)N(C)CC1=C
Drug Score: 1.00, Toxicity: 0.56, Final Score: 0.44
----------------------------------------
```

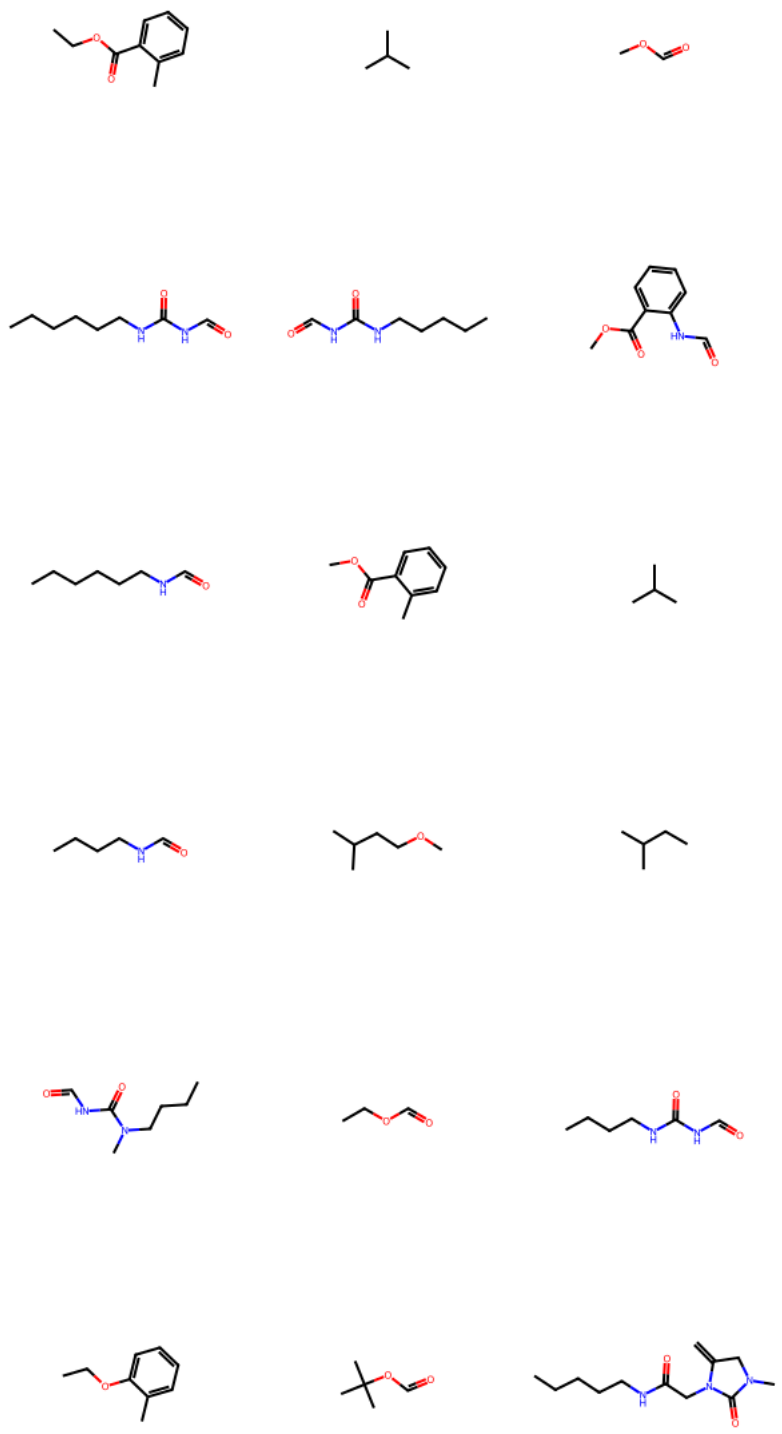
```
results = []

for sm in valid_molecules:
    d = drug_score(sm)
    t = toxicity(sm)
    score = d - t

    results.append((sm, d, t, score))

# Sort by highest score
results = sorted(results, key=lambda x: x[3], reverse=True)

# BEST molecule
best = results[0]

print("\n🏆 BEST MOLECULE:")
print("Molecule:", best[0])
print("Drug Score:", round(best[1], 2))
print("Toxicity:", round(best[2], 2))
print("Final Score:", round(best[3], 2))
```

```
🏆 BEST MOLECULE:
Molecule: CC(C)C
Drug Score: 1.0
Toxicity: 0.12
Final Score: 0.88
```

```
print("\n--- Evaluation Metrics ---")

# 1 Validity
validity = len(valid_raw) / len(generated)
print("Validity:", round(validity, 3))

# 2 Uniqueness
uniqueness = len(set(generated)) / len(generated)
print("Uniqueness:", round(uniqueness, 3))

# 3 Novelty
train_set = set(smiles)   # your training data
novel = [m for m in valid_molecules if m not in train_set]

novelty = len(novel) / len(valid_molecules) if valid_molecules else 0
print("Novelty:", round(novelty, 3))

# 4 Average Drug Score
scores = [drug_score(m) for m in valid_molecules]
avg_score = sum(scores) / len(scores) if scores else 0

print("Avg Drug Score:", round(avg_score, 3))
```

```
--- Evaluation Metrics ---
Validity: 0.073
Uniqueness: 1.0
Novelty: 1.0
Avg Drug Score: 1.0
```

```
from rdkit.Chem import Draw

mols = [Chem.MolFromSmiles(sm) for sm in valid_molecules]
Draw.MolsToGridImage(mols)
```



```
print("\n--- Advanced Metrics (%) ---")

total_valid = len(valid_molecules)

# 1 Drug-like %
drug_like = [m for m in valid_molecules if drug_score(m) >= 0.75]
drug_like_pct = (len(drug_like) / total_valid * 100) if total_valid else 0

# 2 Non-drug-like %
non_drug_like_pct = 100 - drug_like_pct

# 3 Average Toxicity (already a value, keep as is or convert to %)
tox_scores = [toxicity(m) for m in valid_molecules]
avg_toxicity = (sum(tox_scores) / len(tox_scores) * 100) if tox_scores else 0

# 4 Novel %
train_set = set(smiles)
```

```
novel_molecules = [m for m in valid_molecules if m not in train_set]
novel_pct = (len(novel_molecules) / total_valid * 100) if total_valid else 0

print("Drug-Like Molecules (%):", round(drug_like_pct, 2))
print("Non-Drug-Like Molecules (%):", round(non_drug_like_pct, 2))
print("Average Toxicity (%):", round(avg_toxicity, 2))
print("Novel Molecules (%):", round(novel_pct, 2))
```

```
--- Advanced Metrics (%) ---
Drug-Like Molecules (%): 100.0
Non-Drug-Like Molecules (%): 0.0
Average Toxicity (%): 27.22
Novel Molecules (%): 100.0
```

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("basu369victor/zinc250k")

print("Path to dataset files:", path)
```

```
Downloading to /root/.cache/kagglehub/datasets/basu369victor/zinc250k/1.archive...
100%|██████████| 8.60M/8.60M [00:00<00:00, 11.0MB/s]Extracting files...

Path to dataset files: /root/.cache/kagglehub/datasets/basu369victor/zinc250k/versions/1
```

```
!pip install rdkit matplotlib seaborn pandas numpy scikit-learn -q
```

📊 CELL 2 — Full Molecular Property Table (MW, LogP, TPSA, HBA, HBD, Rings, RotBonds)

```
import pandas as pd
from rdkit import Chem
from rdkit.Chem import Descriptors, rdMolDescriptors

rows = []
for sm in valid_molecules:
    mol = Chem.MolFromSmiles(sm)
    if mol is None:
        continue
    rows.append({
        'SMILES'         : sm,
        'MolWeight'      : round(Descriptors.MolWt(mol), 2),
        'LogP'           : round(Descriptors.MolLogP(mol), 2),
        'TPSA'           : round(rdMolDescriptors.CalcTPSA(mol), 2),
        'HBD'            : Descriptors.NumHDonors(mol),
        'HBA'            : Descriptors.NumHAcceptors(mol),
        'RotBonds'       : rdMolDescriptors.CalcNumRotatableBonds(mol),
        'AromaticRings'  : rdMolDescriptors.CalcNumAromaticRings(mol),
        'DrugScore'      : round(drug_score(sm), 3),
        'Toxicity'       : round(toxicity(sm), 3),
        'FinalScore'     : round(drug_score(sm) - toxicity(sm), 3)
    })

prop_df = pd.DataFrame(rows).sort_values('FinalScore', ascending=False).reset_index(drop=True)
print(f'Total valid molecules with properties: {len(prop_df)}')
prop_df.head(15)
```

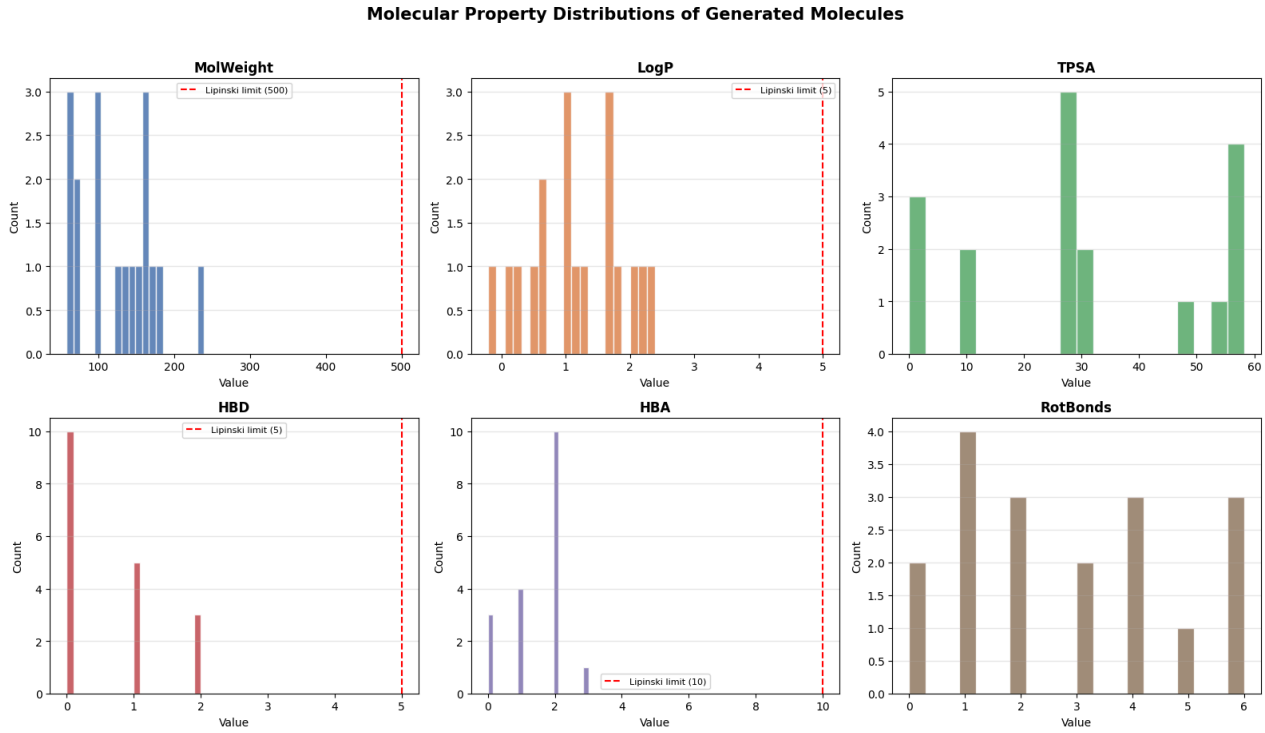
```
Total valid molecules with properties: 18
```

	SMILES	MolWeight	LogP	TPSA	HBD	HBA	RotBonds	AromaticRings	DrugScore	Toxicity	FinalScore
0	CC(C)C	58.12	1.66	0.00	0	0	0	0	1.0	0.12	0.88
1	COC(=O)	60.05	-0.21	26.30	0	2	1	0	1.0	0.14	0.86
2	CC(C)CC	72.15	2.05	0.00	0	0	1	0	1.0	0.14	0.86
3	CC(C)(C)	58.12	1.66	0.00	0	0	0	0	1.0	0.16	0.84
4	CCOC(=O)	74.08	0.18	26.30	0	2	2	0	1.0	0.16	0.84
5	CC(C)CCOC	102.18	1.68	9.23	0	1	3	0	1.0	0.18	0.82
6	CCCCNC(=O)	101.15	0.53	29.10	1	1	4	0	1.0	0.20	0.80
7	CCOc1ccccc1C	136.19	2.39	9.23	0	1	2	1	1.0	0.24	0.76
8	CCCCCCNC(=O)	129.20	1.31	29.10	1	1	6	0	1.0	0.24	0.76
9	CC(C)(C)OC(=O)	102.13	0.96	26.30	0	2	1	0	1.0	0.28	0.72
10	COC(=O)c1ccccc1C	150.18	1.78	26.30	0	2	1	1	1.0	0.32	0.68
11	CCCCNC(=O)NC(=O)	144.17	0.24	58.20	2	2	4	0	1.0	0.32	0.68
12	CCOC(=O)c1ccccc1C	164.20	2.17	26.30	0	2	2	1	1.0	0.34	0.66
13	CCCCCNC(=O)NC(=O)	158.20	0.63	58.20	2	2	5	0	1.0	0.34	0.66
14	CCCCCCNC(=O)NC(=O)	172.23	1.02	58.20	2	2	6	0	1.0	0.36	0.64

Next steps: Generate code with prop_df New interactive sheet

📈 CELL 3 — Property Distribution Plots

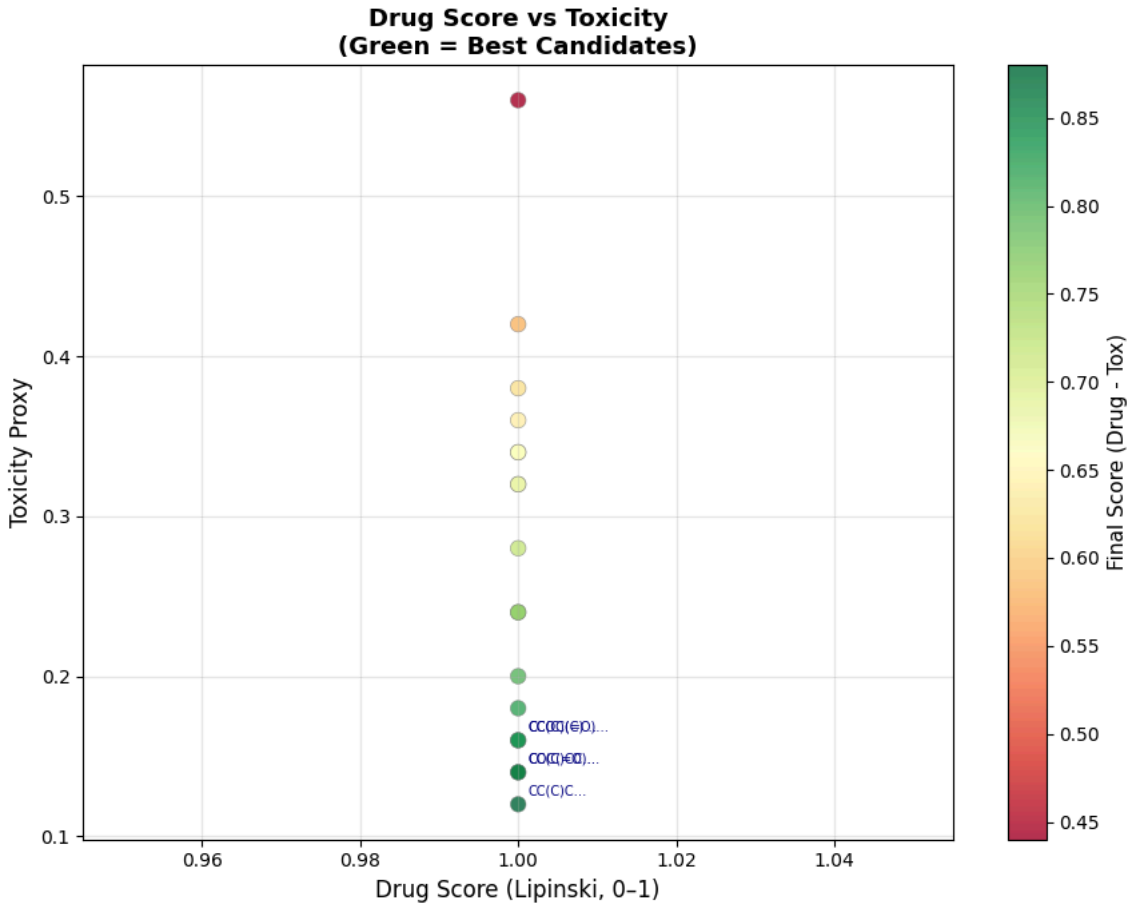
```
import matplotlib.pyplot as plt
import seaborn as sns

props = ['MolWeight', 'LogP', 'TPSA', 'HBD', 'HBA', 'RotBonds']
lipinski_limits = {
    'MolWeight': 500, 'LogP': 5, 'HBD': 5, 'HBA': 10
}

fig, axes = plt.subplots(2, 3, figsize=(16, 9))
axes = axes.flatten()

colors = ['#4C72B0','#DD8452','#55A868','#C44E52','#8172B2','#937860']

for i, prop in enumerate(props):
    ax = axes[i]
    ax.hist(prop_df[prop], bins=20, color=colors[i], edgecolor='white', alpha=0.85)
    if prop in lipinski_limits:
        ax.axvline(lipinski_limits[prop], color='red', linestyle='--',
                   linewidth=1.5, label=f'Lipinski limit ({lipinski_limits[prop]})')
        ax.legend(fontsize=8)
    ax.set_title(prop, fontsize=12, fontweight='bold')
    ax.set_xlabel('Value')
    ax.set_ylabel('Count')
    ax.grid(axis='y', alpha=0.3)

plt.suptitle('Molecular Property Distributions of Generated Molecules',
             fontsize=15, fontweight='bold', y=1.02)
plt.tight_layout()
plt.savefig('property_distributions.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: property_distributions.png')
```




```
Saved: property_distributions.png
```

🎯 CELL 4 — Drug Score vs Toxicity Scatter Plot

```
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

fig, ax = plt.subplots(figsize=(9, 7))

sc = ax.scatter(
    prop_df['DrugScore'],
    prop_df['Toxicity'],
    c=prop_df['FinalScore'],
    cmap='RdYlGn',
    s=70, alpha=0.8, edgecolors='grey', linewidths=0.4
)

cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('Final Score (Drug - Tox)', fontsize=11)

# Annotate top 5
top5 = prop_df.head(5)
for _, row in top5.iterrows():
    ax.annotate(row['SMILES'][:10] + '…',
                xy=(row['DrugScore'], row['Toxicity']),
                xytext=(5, 5), textcoords='offset points',
                fontsize=7, color='navy')

ax.set_xlabel('Drug Score (Lipinski, 0–1)', fontsize=12)
ax.set_ylabel('Toxicity Proxy', fontsize=12)
ax.set_title('Drug Score vs Toxicity\n(Green = Best Candidates)', fontsize=13, fontweight='bold')
ax.grid(alpha=0.3)
plt.tight_layout()
plt.savefig('drug_vs_toxicity.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: drug_vs_toxicity.png')
```




```
Saved: drug_vs_toxicity.png
```

CELL 5 — Chemical Space Map (PCA on Morgan Fingerprints)

```
import numpy as np
import matplotlib.pyplot as plt
from rdkit import Chem
from rdkit.Chem import AllChem
from sklearn.decomposition import PCA

fps, smiles_valid, scores = [], [], []

for _, row in prop_df.iterrows():
    mol = Chem.MolFromSmiles(row['SMILES'])
    if mol is None:
        continue
    fp = AllChem.GetMorganFingerprintAsBitVect(mol, radius=2, nBits=512)
    fps.append(list(fp))
    smiles_valid.append(row['SMILES'])
    scores.append(row['FinalScore'])

X = np.array(fps)

if X.shape[0] >= 2:
    pca = PCA(n_components=2, random_state=42)
    X2d = pca.fit_transform(X)

    fig, ax = plt.subplots(figsize=(10, 7))
    sc = ax.scatter(X2d[:, 0], X2d[:, 1],
                    c=scores, cmap='plasma', s=60,
                    alpha=0.8, edgecolors='white', linewidths=0.3)
    cbar = plt.colorbar(sc, ax=ax)
    cbar.set_label('Final Score', fontsize=11)

    # Star the best molecule
    ax.scatter(X2d[0, 0], X2d[0, 1], marker='*', s=300,
               color='gold', edgecolors='black', zorder=5, label='Best molecule')
    ax.legend(fontsize=10)

    ax.set_xlabel(f'PC1 ({pca.explained_variance_ratio_[0]*100:.1f}% var)', fontsize=11)
    ax.set_ylabel(f'PC2 ({pca.explained_variance_ratio_[1]*100:.1f}% var)', fontsize=11)
    ax.set_title('Chemical Space (PCA on Morgan Fingerprints)', fontsize=13, fontweight='bold')
    ax.grid(alpha=0.3)
    plt.tight_layout()
    plt.savefig('chemical_space_pca.png', dpi=150, bbox_inches='tight')
```

```
    plt.savefig('chemical_space_pca.png', dpi=150, bbox_inches='tight')
    plt.show()
    print('Saved: chemical_space_pca.png')
else:
    print('Not enough valid molecules for PCA (need at least 2).')
```

```
[14:09:29] DEPRECATION WARNING: please use MorganGenerator
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```
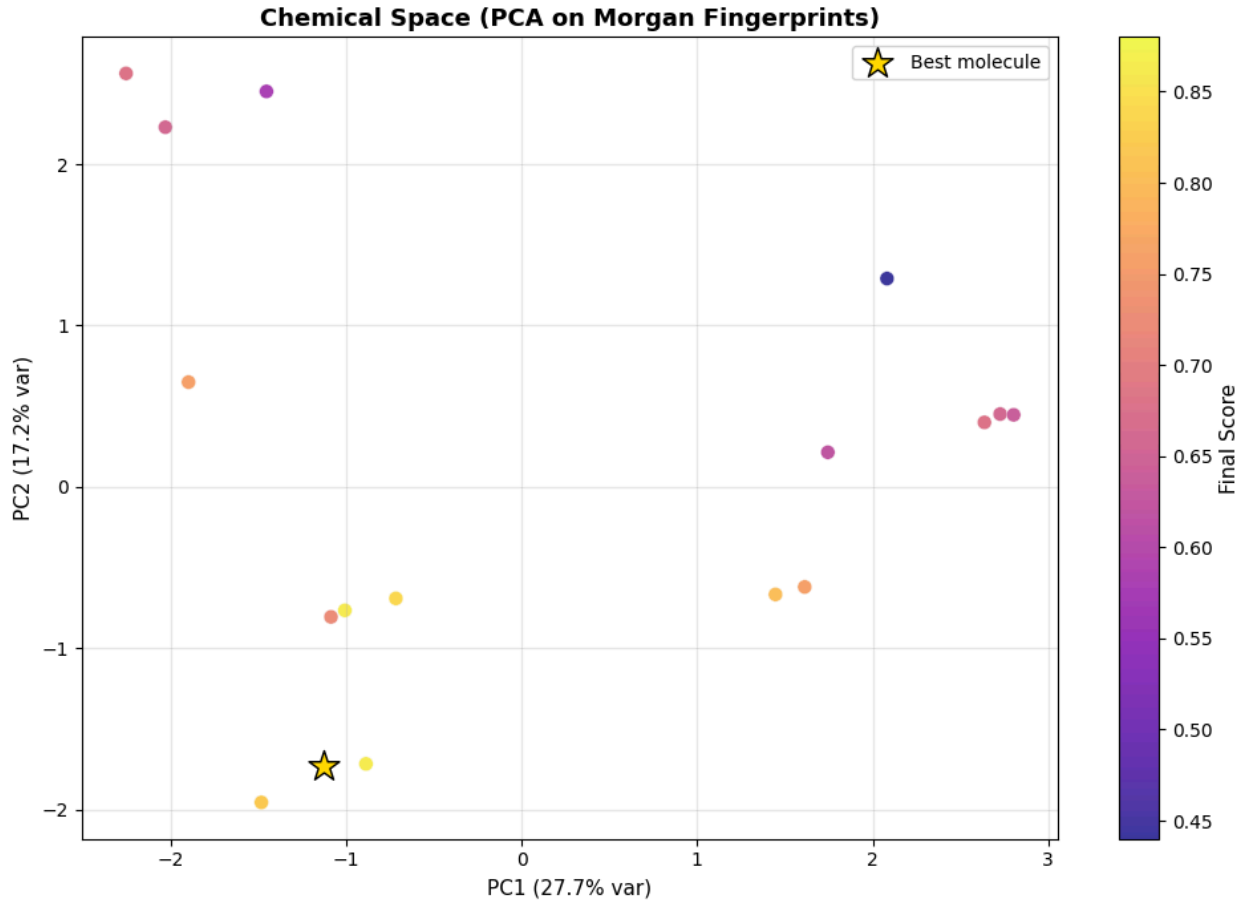
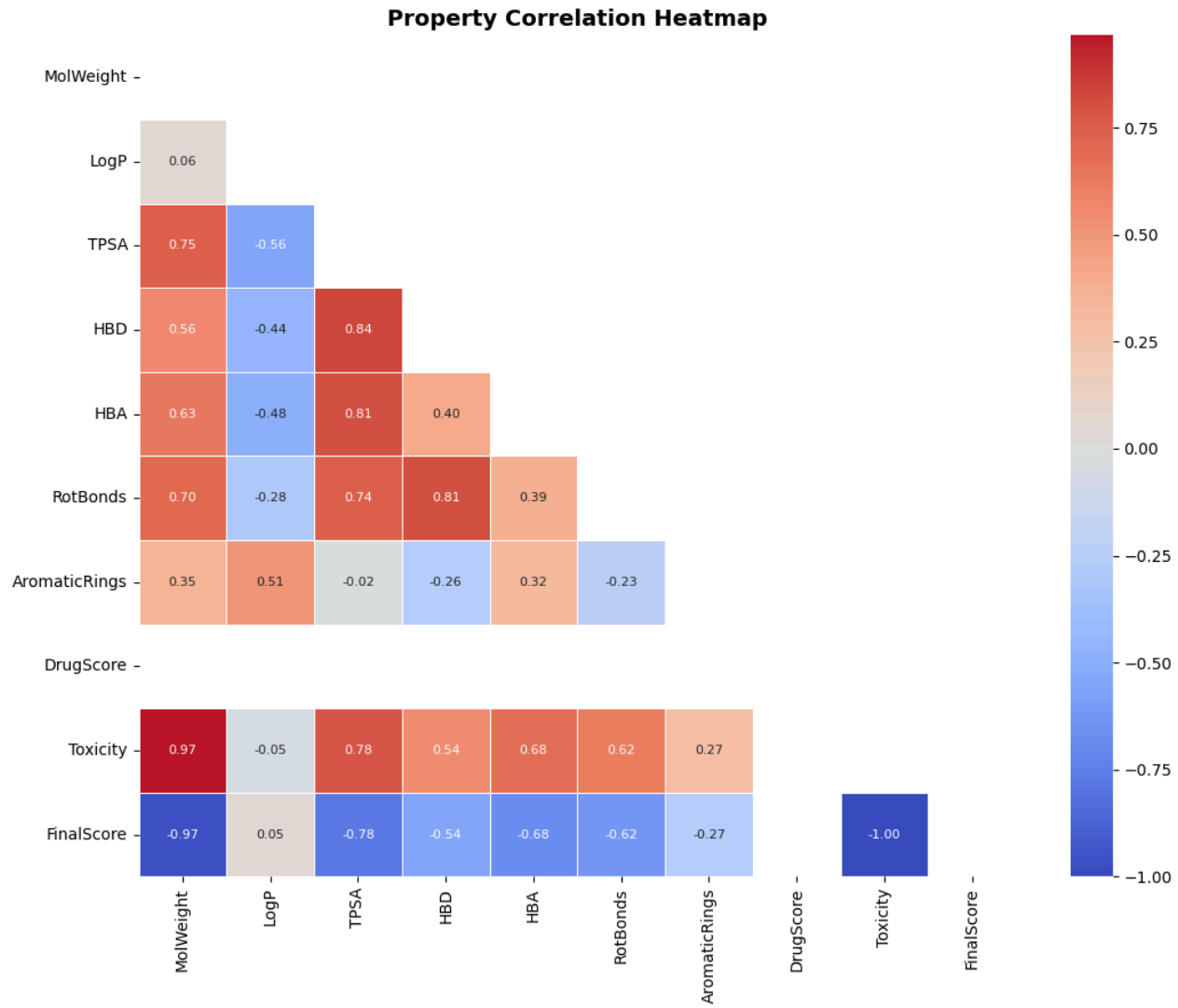




```
Saved: chemical_space_pca.png
```

CELL 6 — Property Correlation Heatmap

```
import seaborn as sns
import matplotlib.pyplot as plt

num_cols = ['MolWeight','LogP','TPSA','HBD','HBA','RotBonds','AromaticRings','DrugScore','Toxicity','FinalScore']
corr = prop_df[num_cols].corr()

fig, ax = plt.subplots(figsize=(11, 9))
mask = np.zeros_like(corr, dtype=bool)
mask[np.triu_indices_from(mask)] = True

sns.heatmap(corr, mask=mask, annot=True, fmt='.2f',
            cmap='coolwarm', center=0,
            linewidths=0.5, ax=ax,
            annot_kws={'size': 8})

ax.set_title('Property Correlation Heatmap', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.savefig('property_correlation.png', dpi=150, bbox_inches='tight')
plt.show()
```

```
print('Saved: property_correlation.png')
```

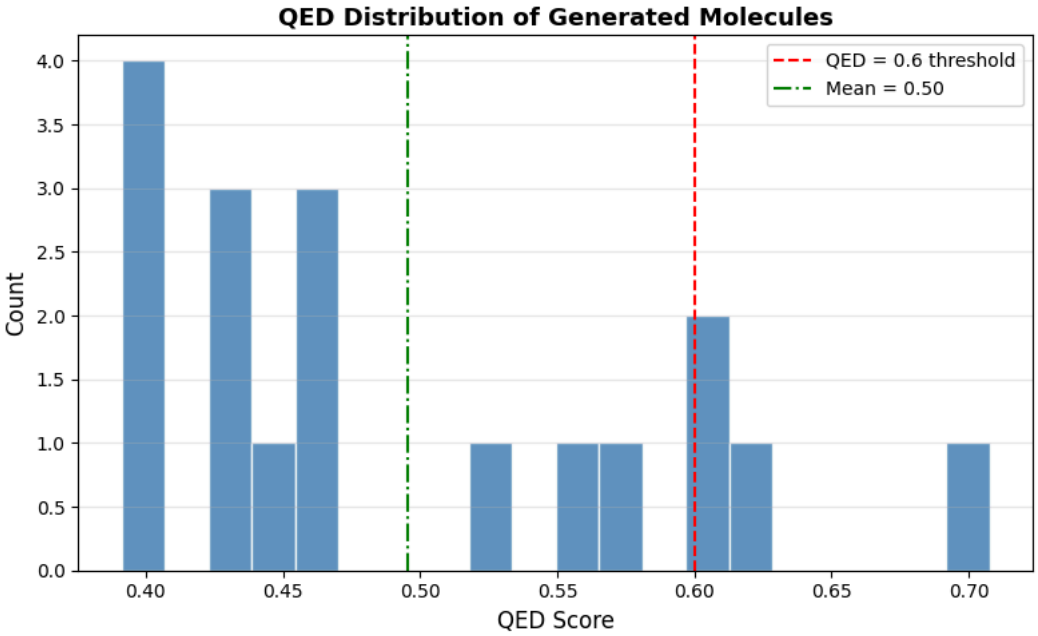



```
Saved: property_correlation.png
```

CELL 7 — QED Score (Quantitative Estimate of Drug-likeness)

```
from rdkit.Chem import QED
import matplotlib.pyplot as plt

qed_scores = []
for sm in valid_molecules:
    mol = Chem.MolFromSmiles(sm)
    if mol:
        qed_scores.append(QED.qed(mol))

print(f'\n--- QED (Quantitative Estimate of Drug-likeness) ---')
print(f'Min QED  : {min(qed_scores):.3f}')
print(f'Max QED  : {max(qed_scores):.3f}')
print(f'Mean QED : {sum(qed_scores)/len(qed_scores):.3f}')
print(f'Molecules with QED > 0.6: {sum(1 for q in qed_scores if q > 0.6)} ({sum(1 for q in qed_scores if q > 0.6)/len(qed_sc

fig, ax = plt.subplots(figsize=(8, 5))
ax.hist(qed_scores, bins=20, color='steelblue', edgecolor='white', alpha=0.85)
ax.axvline(0.6, color='red', linestyle='--', linewidth=1.5, label='QED = 0.6 threshold')
ax.axvline(sum(qed_scores)/len(qed_scores), color='green', linestyle='-.',
           linewidth=1.5, label=f'Mean = {sum(qed_scores)/len(qed_scores):.2f}')
ax.set_xlabel('QED Score', fontsize=12)
ax.set_ylabel('Count', fontsize=12)
ax.set_title('QED Distribution of Generated Molecules', fontsize=13, fontweight='bold')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('qed_distribution.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: qed_distribution.png')
```

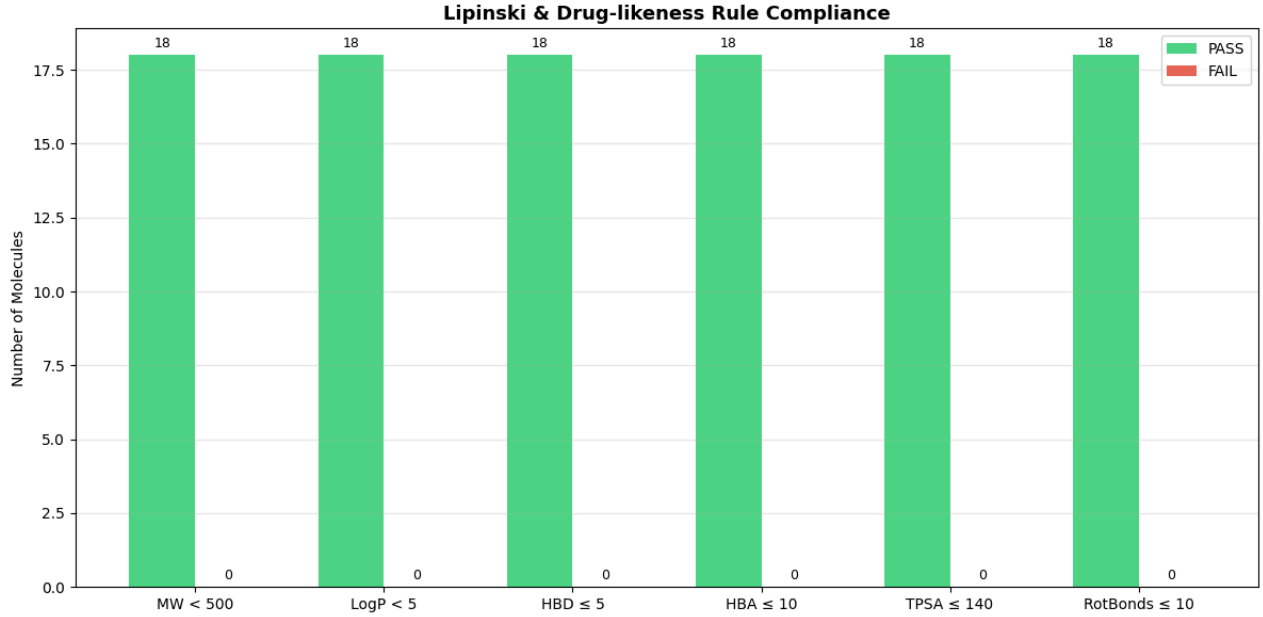
```
--- QED (Quantitative Estimate of Drug-likeness) ---
Min QED  : 0.391
Max QED  : 0.707
Mean QED : 0.495
Molecules with QED > 0.6: 4 (22.2%)
```




```
Saved: qed_distribution.png
```

📉 CELL 8 — Lipinski Rule Violation Breakdown (Bar Chart)

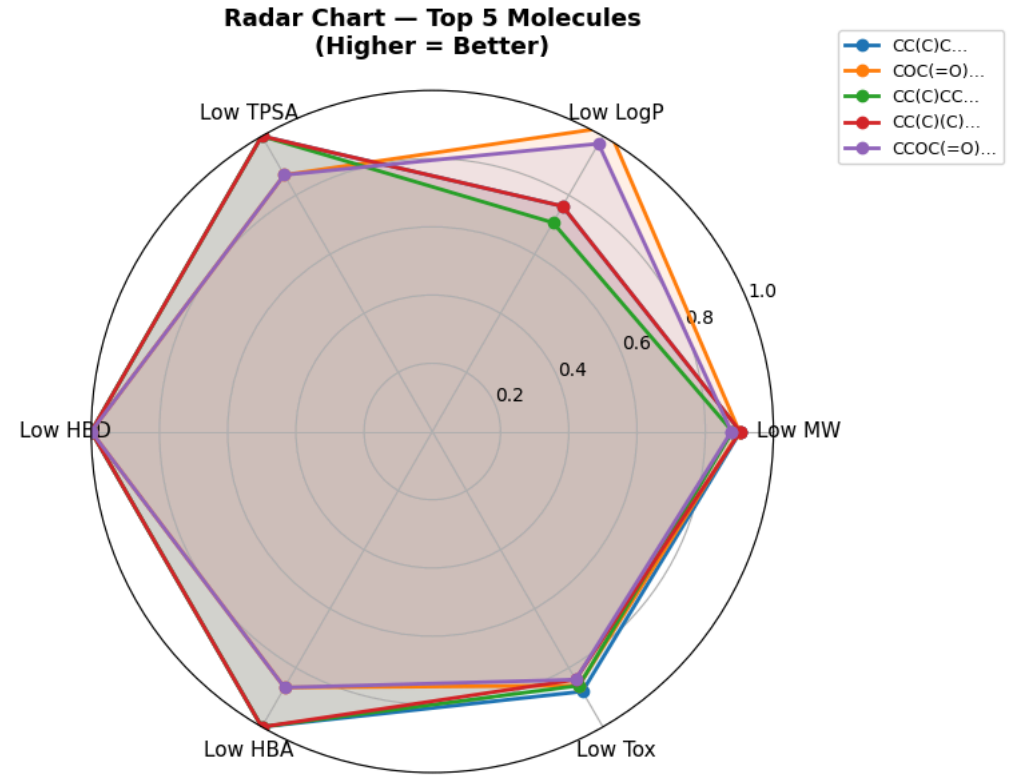
```
import matplotlib.pyplot as plt
import numpy as np

rules = {
    'MW < 500'    : (prop_df['MolWeight'] < 500).sum(),
    'LogP < 5'    : (prop_df['LogP'] < 5).sum(),
    'HBD ≤ 5'     : (prop_df['HBD'] <= 5).sum(),
    'HBA ≤ 10'    : (prop_df['HBA'] <= 10).sum(),
    'TPSA ≤ 140'  : (prop_df['TPSA'] <= 140).sum(),
    'RotBonds ≤ 10': (prop_df['RotBonds'] <= 10).sum(),
}

total = len(prop_df)
labels = list(rules.keys())
pass_counts = list(rules.values())
fail_counts = [total - p for p in pass_counts]

x = np.arange(len(labels))
width = 0.35

fig, ax = plt.subplots(figsize=(12, 6))
bars1 = ax.bar(x - width/2, pass_counts, width, label='PASS', color='#2ecc71', alpha=0.85)
bars2 = ax.bar(x + width/2, fail_counts, width, label='FAIL', color='#e74c3c', alpha=0.85)

ax.bar_label(bars1, padding=3, fontsize=9)
ax.bar_label(bars2, padding=3, fontsize=9)

ax.set_xticks(x)
ax.set_xticklabels(labels, fontsize=10)
ax.set_ylabel('Number of Molecules')
ax.set_title('Lipinski & Drug-likeness Rule Compliance', fontsize=13, fontweight='bold')
ax.legend()
ax.grid(axis='y', alpha=0.3)
plt.tight_layout()
plt.savefig('lipinski_compliance.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: lipinski_compliance.png')
```




```
Saved: lipinski_compliance.png
```

CELL 9 — Radar Chart for Top 5 Molecules

```
import matplotlib.pyplot as plt
import numpy as np

# Normalize properties to [0, 1] for radar (higher = better)
top5 = prop_df.head(5).copy()

def normalize_inv(val, max_val):   # lower raw = higher score
    return max(0, 1 - val / max_val)

radar_data = []
for _, row in top5.iterrows():
    radar_data.append([
        normalize_inv(row['MolWeight'], 600),
        normalize_inv(row['LogP'], 7),
        normalize_inv(row['TPSA'], 200),
        normalize_inv(row['HBD'], 8),
        normalize_inv(row['HBA'], 15),
        normalize_inv(row['Toxicity'], 1),
    ])

categories = ['Low MW', 'Low LogP', 'Low TPSA', 'Low HBD', 'Low HBA', 'Low Tox']
N = len(categories)
angles = [n / float(N) * 2 * np.pi for n in range(N)]
angles += angles[:1]

fig, ax = plt.subplots(figsize=(8, 8), subplot_kw=dict(polar=True))
colors = ['#1f77b4','#ff7f0e','#2ca02c','#d62728','#9467bd']

for i, (values, (_, row)) in enumerate(zip(radar_data, top5.iterrows())):
    vals = values + values[:1]
    ax.plot(angles, vals, 'o-', linewidth=2, color=colors[i],
            label=row['SMILES'][:12] + '…')
    ax.fill(angles, vals, alpha=0.1, color=colors[i])

ax.set_xticks(angles[:-1])
ax.set_xticklabels(categories, fontsize=11)
ax.set_ylim(0, 1)
ax.set_title('Radar Chart — Top 5 Molecules\n(Higher = Better)', fontsize=13, fontweight='bold', pad=20)
ax.legend(loc='upper right', bbox_to_anchor=(1.35, 1.1), fontsize=9)
plt.tight_layout()
plt.savefig('radar_top5.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: radar_top5.png')
```



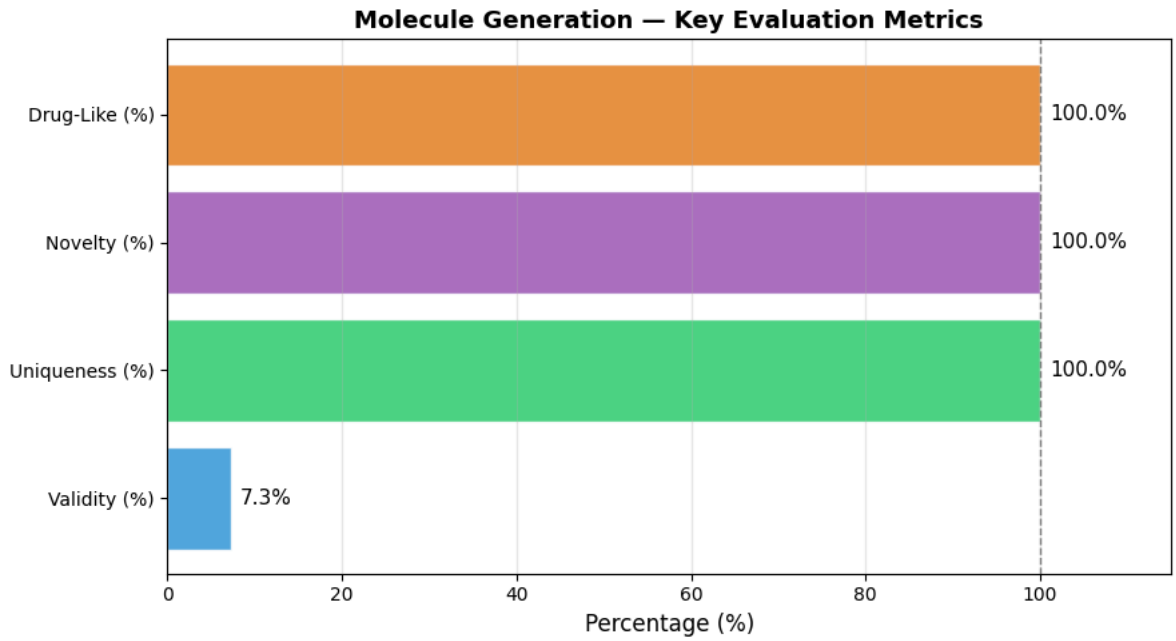

```
Saved: radar_top5.png
```

CELL 10 — Full Metrics Summary Dashboard

```
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec
from rdkit.Chem import QED

# ---- recompute all metrics ----
total_generated = len(generated)
validity_pct    = len(valid_raw) / total_generated * 100
uniqueness_pct  = len(set(generated)) / total_generated * 100
train_set       = set(smiles)
novel_mols      = [m for m in valid_molecules if m not in train_set]
novelty_pct     = len(novel_mols) / len(valid_molecules) * 100 if valid_molecules else 0
drug_like_mols  = [m for m in valid_molecules if drug_score(m) >= 0.75]
drug_like_pct   = len(drug_like_mols) / len(valid_molecules) * 100 if valid_molecules else 0

qeds = []
for sm in valid_molecules:
    mol = Chem.MolFromSmiles(sm)
    if mol:
        qeds.append(QED.qed(mol))
mean_qed = sum(qeds)/len(qeds) if qeds else 0

all_ds = [drug_score(m) for m in valid_molecules]
mean_drug_score = sum(all_ds)/len(all_ds) if all_ds else 0

all_ts = [toxicity(m) for m in valid_molecules]
mean_tox = sum(all_ts)/len(all_ts) if all_ts else 0

metrics = {
    'Validity (%)': round(validity_pct, 1),
    'Uniqueness (%)': round(uniqueness_pct, 1),
    'Novelty (%)': round(novelty_pct, 1),
    'Drug-Like (%)': round(drug_like_pct, 1),
    'Mean QED': round(mean_qed, 3),
    'Mean DrugScore': round(mean_drug_score, 3),
    'Mean Toxicity': round(mean_tox, 3),
    'Total Generated': total_generated,
    'Total Valid': len(valid_molecules),
    'Novel Mols': len(novel_mols),
}

print('\n' + '='*45)
print('        FULL METRICS SUMMARY')
print('='*45)
for k, v in metrics.items():
    print(f'  {k:<20} : {v}')
print('='*45)
```

```
# ---- Dashboard bar chart ----
pct_metrics = ['Validity (%)', 'Uniqueness (%)', 'Novelty (%)', 'Drug-Like (%)']
vals = [metrics[m] for m in pct_metrics]

fig, ax = plt.subplots(figsize=(9, 5))
bar_colors = ['#3498db','#2ecc71','#9b59b6','#e67e22']
bars = ax.barh(pct_metrics, vals, color=bar_colors, alpha=0.85, edgecolor='white')
ax.bar_label(bars, fmt='%.1f%%', padding=5, fontsize=11)
ax.set_xlim(0, 115)
ax.set_xlabel('Percentage (%)', fontsize=12)
ax.set_title('Molecule Generation — Key Evaluation Metrics', fontsize=13, fontweight='bold')
ax.axvline(100, color='grey', linestyle='--', linewidth=1)
ax.grid(axis='x', alpha=0.3)
plt.tight_layout()
plt.savefig('metrics_dashboard.png', dpi=150, bbox_inches='tight')
plt.show()
print('Saved: metrics_dashboard.png')
```

```
==============================================
        FULL METRICS SUMMARY
==============================================
  Validity (%)         : 7.3
  Uniqueness (%)       : 100.0
  Novelty (%)          : 100.0
  Drug-Like (%)        : 100.0
  Mean QED             : 0.495
  Mean DrugScore       : 1.0
  Mean Toxicity        : 0.272
  Total Generated      : 245
  Total Valid          : 18
  Novel Mols           : 18
==============================================
```




```
Saved: metrics_dashboard.png
```

🖼 CELL 11 — Visualize Top 12 Molecules (RDKit Grid Image)

```
from rdkit import Chem
from rdkit.Chem import Draw
from IPython.display import display, Image # Import Image directly
import io

# Get top 12 rows from prop_df
top_rows = prop_df.head(12)

top12_mols = []
legends = []
for index, row in top_rows.iterrows():
    sm = row['SMILES']
    mol = Chem.MolFromSmiles(sm)
    if mol is not None:
        top12_mols.append(mol)
        legends.append(f'#{len(top12_mols)} Score:{row["FinalScore"]:.2f}')
```